Sprint 3.1 — Baseline State & Lever Catalog

Deloitte Capstone: AI for Equitable Public Transportation
April 9, 2026

What Sprint 3.1 Produced

- **Sprint3_Baseline_State.csv** — 504 tracts x 29 columns. Single file the simulator loads at startup. Contains the 13 XGBoost v3 features, deficit predictions, need projections, equity tiers, and fragile/worsening flags. Built from a 3-way inner join of the features matrix, v3 risk scores, and ACS time-series projections.
- **Sprint3_Lever_Catalog.json** — 4 transit-service levers with SHAP importance ranks, valid delta ranges, and data ranges. Includes design rationale and future-feature notes.
- **Sprint 3/future/** — Quarantined OLS need coefficients (Daniel's work preserved for potential v2 stress-test feature).

Lever Catalog: Why Only 4 Transit-Service Levers

Daniel's original catalog had 8 levers: 6 access-side (including headway duplicates) and 2 need-side (poverty rate, no-vehicle households). We trimmed to 4 for three reasons:

1. Headway and frequency are physically inverse

- headway_min = 60 / freq_tph. Exposing both as independent levers allows physically impossible states (e.g., low headway AND low frequency simultaneously).
- Fix: expose only frequency to the user. The simulator auto-derives headway = 60 / freq internally before passing all 13 features to XGBoost.
- Frequency is the better user-facing lever: SHAP #1 (39.6%) and #3 (6.9%) vs headway at #6 (3.8%), and more intuitive for planners ('add 2 trips/hour' vs 'cut 8 minutes off headway').

2. Demographic variables are not policy levers

- poverty_rate_pct and hh_no_vehicle_pct are forecast outputs from the ACS time-series model, not parameters a transit agency can control.
- A simulator lever must be something the decision-maker can change by signing a work order. Poverty and vehicle ownership fail this test.
- These variables flow into projected_need_2027 as fixed baseline inputs.

3. Stress-test scenarios deferred to v2

- Demographic shock scenarios ('what if poverty rises 5pp in fragile tracts?') are a legitimate future feature, but architecturally distinct from policy levers.
- If re-enabled in v2, they should be labeled 'stress tests' in the UI (not 'levers') and use the Sprint 2a composite formula weights directly rather than OLS regression (which has a circularity issue — see below).
- Daniel's OLS coefficients file is preserved in Sprint 3/future/ with a README documenting the deferral rationale.

Final 4 Levers

Lever	SHAP Rank	Unit	Delta Range	Baseline Mean
freq_peak_am_tph	#1 (39.6%)	trips/hr	[-10, +10]	26.7
weekend_weekday_ratio	#2 (21.7%)	ratio	[-0.5, +0.5]	0.43
freq_early_tph	#3 (6.9%)	trips/hr	[-5, +5]	2.98
rail_trip_share	Modal lever	fraction	[-0.1, +0.3]	0.011

Note: rail_trip_share is not in the SHAP top 6 but is the single most impactful lever in what-if analysis: +10pp produces -0.0149 deficit reduction with 98% of tracts improving — roughly 2x the next best lever.

OLS Need Coefficients — Why Quarantined

- Daniel built an OLS model (R^2=0.817) regressing composite_need on poverty_rate_pct and hh_no_vehicle_pct. Purpose: local linearization for need-side stress tests.
- **Circularity concern:** composite_need was constructed from these same features in Sprint 2a. The OLS is partially fitting a formula to its own components — same pattern as Lesson 16 (the original V1 circular model).
- **Not needed in v1:** the simulator only exposes transit-service parameters. Need side flows from the ACS forecast as a fixed baseline input.
- **If re-enabled:** use the Sprint 2a composite formula weights directly instead of OLS regression. More transparent and avoids the circularity.
- File preserved at Sprint 3/future/Sprint3_Need_OLS_Coefficients.json with a README.

Scale Mismatch Resolution — How the Simulator Computes Equity

The problem

- Sprint 2a equity formula: equity_priority_score = composite_need (0-1) x composite_access_deficit (0-1).
- The v3 model predicts service_deficit (5-indicator index, 0-1 scale) — a **different construct** from composite_access_deficit (7-indicator composite). Correlation is only 0.61.
- The ACS time-series produces projected_need_2027 (0-100 scale) — a **different construct** from composite_need (0-1 scale). Rank correlation is only 0.67.
- Naively multiplying any combination of these mismatched measures makes the Sprint 2a tier cutoffs meaningless. Testing confirmed: naive approach recovers only 57% of current tier assignments at zero delta.

The solution: proportional-change approach

- The simulator uses XGBoost v3 to compute a **ratio of change**, not a replacement score:

```
ratio = deficit_predicted_new / deficit_predicted_baseline
```

```
new_equity = composite_need x (composite_access_deficit x ratio)
```

- At zero delta, ratio = 1 and the equity score is exactly equity_priority_score. Tier recovery: **99.6%**.
- Sprint 2a tier cutoffs remain valid because we operate on the Sprint 2a scale.

- XGBoost v3 is used for what it's good at: predicting direction and magnitude of deficit change from transit features.
- projected_need_2027 and fragile/worsening flags stay in the baseline for **filtering and context** (e.g., 'show me only fragile tracts'), but do NOT enter the equity multiplication.

Zero-Frequency Tracts

- 176 tracts (35%) have freq_peak_am_tph = 0 (no peak AM transit service). 209 tracts (41%) have freq_early_tph = 0.
- In the source data, all zero-frequency tracts have headway set to 120 min (the GTFS pipeline's 'no service' cap). XGBoost was trained on this convention.
- Since the simulator derives headway = 60 / freq, zero frequency causes division by zero.
- **Fix:** when post-delta frequency is at or below 0, cap headway at 120 min (matching training data). Clamp frequency to 0 (no negative trips/hr).
- **Note for interpretation:** adding service to a zero-frequency tract (e.g., +2 tph) creates a large feature swing (headway: 120 min to 30 min). This is correct — going from no service to some service IS a big change — but the simulator should flag these tracts in output so users understand why they show outsized improvement.

Baseline State Assembly — Data Notes

- **Three source files joined on tract_geoid (inner join, 504 tracts):**

Source File	Key Columns Pulled	Rows
Sprint2b_Modeling_Features_NotebookOutput.csv	13 v3 model features + equity_priority_score, equity_tier, composite_need, composite_access_deficit	504
Sprint2b_Risk_Scores_v3.csv	deficit_actual, deficit_predicted, residual, risk_tier	504
Sprint2b_Tract_Risk_Projections.csv	projected_need_2027, proj_pov_2027, proj_veh_2027, flag_worsening, flag_fragile, tract_status	710*

**710 rows cover full county ACS; inner join reduces to 504 (MPO planning area tracts with complete features). Tract 12086009050 was duplicated 4x in the projections file (identical rows) — deduped before join.*

- Sprint2b_Combined_Risk_Scoring.csv (referenced in earlier plans) was never produced. The baseline state CSV absorbs that role — it IS the combined join of deficit + need, plus the model features.
- Verification: all values trace exactly to source files (zero diff), equity formula confirmed (need x deficit = equity score), zero nulls in all model columns, tier cutoffs are monotonically ordered.

What's Next

- **Sprint 3.2 (Apr 10-13):** Build simulator.py — clean importable API that loads the baseline state, accepts lever deltas, re-predicts deficit via XGBoost v3, applies proportional-change equity update, outputs before/after comparison with tier shifts.
- **Sprint 3.3 (Apr 14-16):** Scenario validation notebook — 5 reference scenarios (peak freq boost, weekend parity, early service, rail modal shift, combined multi-lever). Validate against SHAP rankings and 2b.5 what-if results.